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Investigation and Design of an EMG Measurement System (Part 1)

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Abstract: This report presents the development of an EMG signal measurement device designed for upper limb prosthetic control. The objective was to design a non-invasive and cost-effective system capable of capturing and processing surface EMG signals. These signals required high signal fidelity for downstream prosthetic control mechanisms. A structured design approach was followed, involving the selection of dry electrodes, differential amplification, filtering, and microcontroller integration for the proposed solution. Design decisions were guided by signal range requirements, noise performance, and device durability. The proposed solution was evaluated against comparable alternatives using literature and performance criteria, demonstrating a balance between usability, signal quality and system integration potential. Recommendations are for further improvements in noise reduction and ergonomic design.

Keywords: Measurement system, EMG electrodes, myoelectric control, bionic hand, prosthetic control

1. INTRODUCTION

The loss of an individual's limb can significantly hinder their ability to perform everyday tasks. However, through prosthesis one can gain back some of the crucial functionality once lost. These robotic prosthetics work by measuring electromyography (EMG) signals that are generated by the remaining muscles contractions [1]. These EMG signals contain information about muscle performance, fatigue, and control mechanisms [2]. Therefore, creating an accurate and consistent EMG measuring system is a crucial component of the prostatic system, such as that of a bionic hand. The design and analysis of a dry surface EMG measurement device is detailed in this report. A detailed description of the design, application and basic use is provided. This is followed by an analysis of the expected EMG signals. The specifications of the system are then detailed, along with a review of existing systems. A high-level solution is proposed, followed by a comparison of available sensors and concluding the report with future recommendations.

2. DESCRIPTION OF MEASUREMENT APPLICATION

Muscle movement is governed by the interaction between the nervous system and skeletal muscles, where neuromuscular junctions (NMJs) transmit signals from motor neurons to muscle fibres. According to Putcha, an action potential (AP) triggers depolarisation which propagates along the sarcolemma, resulting in muscle contraction [2]. After contraction, repolarisation occurs restoring the resting potential via ion exchange, generating EMG signals as a summative electrical response of the combined AP [2, 3].

These EMG signals arise from voltage fluctuations which can be detected using either surface electrodes (sEMG) or intramuscular electrodes (imEMG) [4, 5]. While imEMG offers better noise immunity and signal strength, it is invasive. Thus, making sEMG the preferred choice despite varying performance across different electrode material and tissue interference, where sliver chloride electrodes are preferred for this application due to their signal conduction properties [2, 6].

Maintaining high EMG signal fidelity is crucial for prosthetic applications for ease of use. Dr De Luca notes the use of bipolar detection through the use of differential amplifiers as it reduces muscle crosstalk noise and selectively amplifies low-amplitude signals [1]. Without differential amplification, both noise and signal are amplified equally. It is further noted by Putcha that a high-input impedance and high common-mode rejection ratio (CMRR) are necessary for minimising power-line interference and improving the signal-to-noise ratio (SNR) [1, 2]. The resulting raw EMG signal contains unwanted frequencies, artefacts and noise. In order to remove the noise, the signal should be passed through analogue filters, being that analogue are more power and cost effective than digital filters. A high-pass filter (10 Hz) removes DC components, while a low-pass filter (500 Hz) eliminates high-frequency noise. Lastly, A notch filter removes the 50 Hz power-line interference [1, 2, 6].

After filtering the signal can be sampled (Nyquist sampling) and digitally processed. Wavelet transformation and machine learning models are tools that can be used to analyse the signal for motor control. Particularly, machine learning can be used to identify individual motor unit firing patterns, offering greater prosthetic control [2, 7].

Figure 1 depicts the proposed measurement system and describes the positioning and basic use [8, 9]. This was designed with practical constraints and assumptions. Namely, the system should be non-invasive and operate in an environment with signal interference like that of power lines and motion artifacts. Also, it should have real-time control, high precision, high resolution, operate linearly, have stable and consistent performance. Lastly the system should be durable for everyday use.

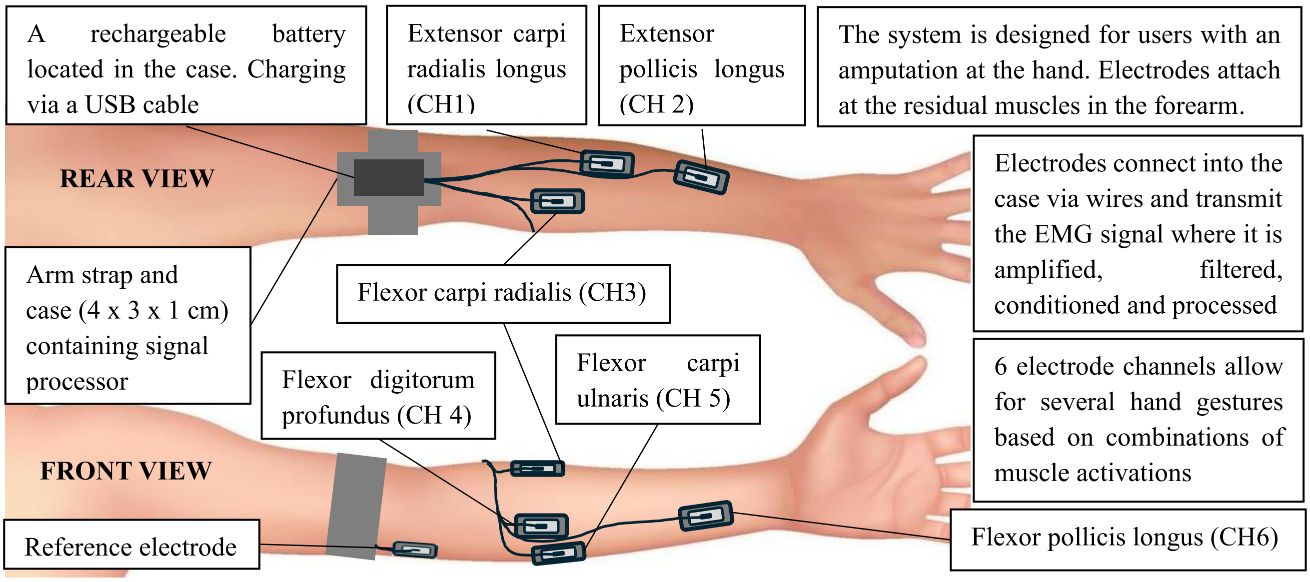

Figure 1: Diagram showing the positioning of the components of the measurement system and basic use [8,9]

3. ANALYSIS OF MEASURED VARIABLE

The measured variable in the system is the EMG signal produced in the arm's muscle as a result of muscle contractions. These EMG signals appear as a burst of electrical activity as seen by figure 2(a). These EMG signals have a range of 0 to 10 mV when measured by a surface electrode [1, 10, 11]. These signals are a summative signal of multiple superimposed motor unit action potentials (MUAP) as seen by figure 2(c). Figure 2(a) and (b) had been derived raw EMG senor data [12].

The sEMG signal can be decomposed into its comprising frequencies ranging from 10 to 500 Hz where most of the signal power lying between 20 and 150 Hz as seen by figure 2(b). The mean value of the frequency of sEMG signals ranges between 40 Hz and 150 Hz, while the median frequency falls around 60 Hz to 100 Hz [10]. Doud noted that muscle fatigue can affect the frequency domain by causing a downward shift in median frequency, showing a reduction in muscle force [13].

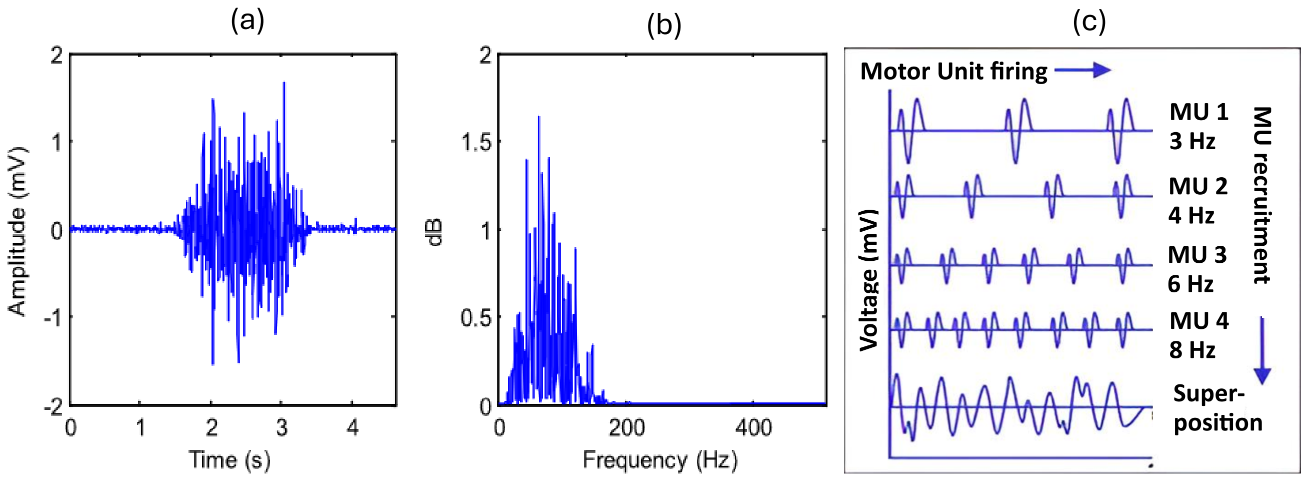

Figure 2: The expected signals into the electrodes. (a) and (b) are, respectively, the time and frequency EMG data. (c) is the decomposition of the EMG signal into individual MUAPs across time [11,13]

4. DESIGN SPECIFICATION

Table 1 details the design specifications and justification for the measurement systems required minimum specifications.

Table 1: Specifications for the designed EMG measurement system and the justifications for each value

Specification	Value	Justification
Input range	0 – 10 mV	Typical EMG amplitude [10, 14]
Bandwidth	20–500 Hz	Covers EMG signal frequencies [2]
Sensor accuracy	±5 µV	Ensures detectable signal variations [1]
Amplifier noise rejection	> 80 dB CMRR	Reduces common-mode interference [2, 12]
Power consumption	<100 mW	Long device operation battery life with small battery
Microprocessor ADC	>1 KHz, 12 bit	Reducing signal aliasing and high resolution
Sensor	sEMG	Non-invasive and improved SNR [1, 2]
Environmental protection	IP54	Dust and water splash resistant [15]

5. REVIEW OF EXISTING SOLUTIONS

Comparisons of existing solutions are found in *Table 2*

Table 2: Comparison of existing solutions

System	Electrode Type	Wireless	Pros	Cons
OpenBCI Ganglion [16]	Wet (Ag/AgCl)	Yes	Open-source, Affordable	Requires gel, few channels
Delsys Trigno[17]	Hybrid (Dry/Wet)	Yes	High accuracy, multi-sensor	Expensive, Complex setup

6. PROPOSED HIGH-LEVEL SOLUTION

The proposed high-level solution a measurement system for EMG signals, as seen by Bentley's model in figure 3, depicts 5 key stages. The design is non-invasive, reliable and durable for prosthetic applications. Existing systems were evaluated to ensure that this design balances signal fidelity, cost-effectiveness, and ease of integration into the bionic hand's control systems.

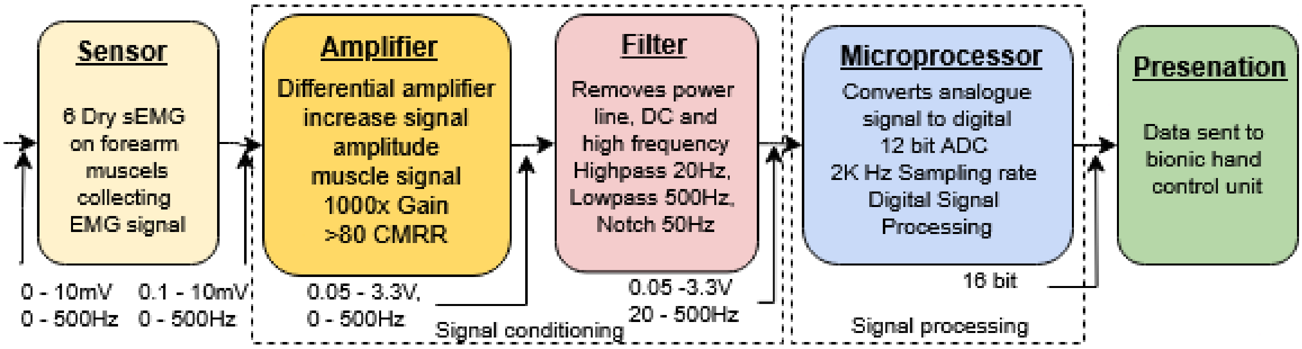

Figure 3: Bentley's model of the EMG measurement system

7. SENSOR SELECTION

A selection of several electrodes is compared in *Table 3* where the GSR/GSC electrodes are found to be the optimal choice for this application. The non-invasiveness, reusability, simple application and long-life span making it a simple and cost-effective choice. However, small compromises are made on SNR which can be limited through secure fastening and signal processing.

Table 3: Comparisons of electrode specifications

Category	**Kendall - Gel sEMG** [5, 18]	**UltraPoint - Needle imEMG** [4, 19, 20]	**Neurotrode - Dry sEMG** [5, 11, 21]	**GSR/GSC – Dry sEMG** [22]
Operating Range	0–10 mV 0–500 Hz	0–30 mV 0–10 kHz	0–10 mV 0–500 Hz	0–10 mV 0–500 Hz
Invasiveness	Non-invasive	Invasive	Non-invasive	Non-invasive
Material	Ag/AgCI	Stainless-steel	Ag/AgCI	Ag/AgCI
Application Complexity	skin and gel preparation	Requires expert insertion	Easy to apply, skin contact	Easy to apply, skin contact
Signal Noise	Low – Conductive gel improves SNR	Very Low, Direct muscle fibre contact	Moderate, weak to motion artifacts	Moderate, weak to motion artifacts
Life Expectancy	Single use (gel dries)	degrades over time	Disposable, non-reusable	Long-term use, reusable
Cost	$0.44 per electrode	$3.24 per electrode	$1.00 per electrode	$0.31 per electrode

8. SUMMARY AND CONCLUSIONS

The measurement system successfully meets the requirements initially set out for reliable measurement of EMG signals. The system offers a trade-off between signal clarity and ease of use where dry electrodes were justified for their balance of reusability and performance. The system's modular design supports future integration with prosthetic control units. The proposed solution is a viable design for real-time EMG signal interpretation in wearable applications, with future work aimed at improving noise immunity and user comfort.

REFERENCES

1. De Luca, C.J.: The Use of Surface Electromyography in Biomechanics. Journal of Applied Biomechanics. 13, 135–163 (1997). https://doi.org/10.1123/jab.13.2.135
2. Putcha, A., Nguyen, T., Smith, R., Choffin, R., Bai, W.: Intelligent Systems for Muscle Tracking: A Review on Sensor-Algorithm Synergy. Advanced Intelligent Systems. 5, 2200351 (2023). https://doi.org/10.1002/aisy.202200351
3. Farina, D., Merletti, R., Enoka, R.M.: The extraction of neural strategies from the surface EMG: an update. Journal of Applied Physiology. 117, 1215–1230 (2014). https://doi.org/10.1152/japplphysiol.00162.2014
4. Rubin, D.I.: Needle electromyography: Basic concepts. Handb Clin Neurol. 160, 243–256 (2019). https://doi.org/10.1016/B978-0-444-64032-1.00016-3
5. Papandrea, P.: Dry vs. Gelled Electrodes in sEMG: Pros and Cons, https://www.cometasystems.com/dry-vs-gelled-electrodes-in-surface-emg-advantages-and-disadvantages/, (2024)
6. Muceli, S., Merletti, R.: Tutorial. Frequency analysis of the surface EMG signal: Best practices. Journal of Electromyography and Kinesiology. 79, 102937 (2024). https://doi.org/10.1016/j.jelekin.2024.102937
7. Wu, D., Tian, P., Zhang, S., Wang, Q., Yu, K., Wang, Y., Gao, Z., Huang, L., Li, X., Zhai, X., Tian, M., Huang, C., Zhang, H., Zhang, J.: A Surface Electromyography (sEMG) System Applied for Grip Force Monitoring. Sensors. 24, 3818 (2024). https://doi.org/10.3390/s24123818
8. Healthy human arm of the Asian people., https://www.vecteezy.com/vector-art/9485254-healthy-human-arm-of-the-asian-people-front-and-back-view-for-advertising-medical-publications-fitness-and-health-concept-on-a-white-background-3d-vector-eps10
9. Jiralerspong, T., Nakanishi, E., Liu, C., Ishikawa, J.: Experimental Study of Real-Time Classification of 17 Voluntary Movements for Multi-Degree Myoelectric Prosthetic Hand. Applied Sciences. 7, 1163 (2017). https://doi.org/10.3390/app7111163
10. Wang, Y., Tian, Y., Zhu, J., She, H., Yokoi, H., Jiang, Y., Huang, Q.: A Study on the Classification Effect of sEMG Signals in Different Vibration Environments Based on the LDA Algorithm. Sensors (Basel). 21, 6234 (2021). https://doi.org/10.3390/s21186234
11. Ismail, R.: Muscle Power Signal Acquisition Monitoring Using Surface EMG. J Biomed Res Environ Sci. 3, 663–667 (2022). https://doi.org/10.37871/jbres1493
12. Wu, C., Zeng, H., Song, A., Xu, B.: Grip Force and 3D Push-Pull Force Estimation Based on sEMG and GRNN. Front. Neurosci. 11, (2017). https://doi.org/10.3389/fnins.2017.00343
13. Doud, J.R., Walsh, J.M.: Muscle fatigue and muscle length interaction: effect on the EMG frequency components. Electromyogr Clin Neurophysiol. 35, 331–339 (1995)
14. TMSi — an Artinis company — What Is Electromyography (EMG)?, https://www.tmsi.artinis.com/blog/what-is-emg
15. He, C.: Guide to IP54 Enclosures: Protection Levels & Standards - E-Abel, https://www.eabel.com/guide-to-ip54-enclosures-protection-levels-standards/, (2025)
16. Ganglion Board (4-channels), https://shop.openbci.com/products/ganglion-board
17. Trigno® Avanti Sensor - Delsys, https://delsys.com/trigno-avanti/
18. Kendall Disposable Surface EMG/ECG/EKG Electrodes 1" (24mm) 50pkg, https://bio-medical.com/covidien-kendall-disposable-surface-emg-ecg-ekg-electrodes-1-24mm-50pkg.html
19. Tankisi, H., Burke, D., Cui, L., de Carvalho, M., Kuwabara, S., Nandedkar, S.D., Rutkove, S., Stålberg, E., van Putten, M.J.A.M., Fuglsang-Frederiksen, A.: Standards of instrumentation of EMG. Clinical Neurophysiology. 131, 243–258 (2020). https://doi.org/10.1016/j.clinph.2019.07.025
20. UltraPoint™ Monopolar EMG Needle - Variants, https://rochestersuperstore.com/ultrapoint-monopolar-emg-needle---variants/

21. Neurotrode Dry EEG/EMG/ECG Sensors - 50 pack, https://bio-medical.com/neurotrode-dry-eeg-emg-ecg-sensors-50-pack.html
22. GSR/GSC Dry Electrodes - MindWare Technologies, https://www.mindwaretech.com/shop/gsr-gsc-dry-electrodes/